

Benjamin Farsijani

Ottawa, ON | 343-262-4264 | farsijaniben@gmail.com | benfarsi.com | github.com/benfarsi | linkedin.com/in/benfarsi

EDUCATION

University of Ottawa

Sep 2023 – May 2028

Honours B.Sc., Computer Science — GPA: 3.7 / 4.0

Ottawa, ON

Relevant Coursework: Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, Systems Programming

TECHNICAL SKILLS

Languages: Python, Go, C/C++, Java, TypeScript/JavaScript, SQL, Bash

ML & Data: PyTorch, XGBoost, scikit-learn, NumPy, pandas; RAG pipelines, hybrid retrieval, pgvector, time-series modeling, anomaly detection

Systems & Infra: Linux, multithreading, TCP/IP, Docker, PostgreSQL, SQLite, Flask, REST/WebSocket

Embedded: ESP32 firmware, I2C/SPI/UART, sensor integration

PROJECTS

Sage — Agentic RAG System for Technical Documentation

github.com/benfarsi/sage

Python, LangChain, pgvector, Claude API, PostgreSQL, Docker

- Built a production-grade agentic RAG system that ingests large-scale networking documentation and answers natural-language queries over complex technical content.
- Engineered hybrid retrieval combining dense semantic search (pgvector + text-embedding-3-large) with BM25 sparse retrieval, merged via reciprocal rank fusion; sub-100ms p95 latency over 100k+ embedded chunks.
- Designed a multi-step agentic loop with tool use that dynamically re-queries and synthesizes across document chunks before generating a final grounded answer; fully containerized ingestion and query pipelines in Docker.

Benny — Algorithmic Crypto Trading System

github.com/benfarsi/benny

Python, XGBoost, CCXT, Binance API, PostgreSQL, Docker, Flask

- Built an end-to-end ML trading system ingesting real-time Binance data across 6 liquid pairs, running continuously with a containerized Flask dashboard tracking live equity and trade history.
- Engineered 50+ zero-lookahead features (GARCH(1,1) volatility, Parkinson range, VWAP deviation, OBV, ADX, Stochastic RSI) feeding an XGBoost directional classifier validated via 5-fold walk-forward CV.
- Implemented a full risk layer: probability-gated entries, 3% take-profit / 1.5% stop-loss, fee modeling, and per-symbol sizing across a \$6K paper portfolio.

Celsius — IoT Environmental Monitoring System (1st Place, Design Day)

github.com/benfarsi/celsius

ESP32, C/C++, React Native, BME680, MAX9814, ML

- Designed and deployed a wall-mounted IoT monitoring system for Garderie Bernadette Child Care Centre at uOttawa, winning 1st place out of 30+ teams; running continuously at the site since launch.
- Integrated BME680 (temp/humidity/VOC) and MAX9814 (sound) over I2C with WiFi streaming, a 3D-printed battery-powered enclosure, a React Native companion app, and an ML anomaly-detection layer with real-time push alerts.

EXPERIENCE

Founder — Benny Blends Barbershop

2018 – Present

- Founded and operated a barbershop from a home studio while enrolled full-time; \$25,000+ cumulative revenue with thousands of clients served, running all operations independently.
- Managed scheduling, inventory, pricing, and marketing, building a consistent referral base with zero advertising spend.

Technician — TPH Plumbing and Heating

Aug 2025 – Feb 2026

- Worked on large commercial job sites (crews of 10+, buildings up to 14 stories) alongside licensed plumbers and gas technicians: cut, threaded, soldered, and fit pipe and coordinated daily with electricians, HVAC, and GCs to keep multi-trade work on schedule.

Competitive Soccer Coach — Ottawa South United

Mar 2019 – July 2025

- Coached 500+ youth athletes across competitive and recreational programs over five seasons, leading training sessions, game-day strategy, and player development.